Taner Cokyasar

9700 S. Cass Ave. Lemont, IL 60439
tcokyasar@anl.gov http://tanercokyasar.com/ (706) 308-1492
0000-0001-9687-6725 Scholar taner-cokyasar

Education

***The University of Tennessee*, Knoxville, TN, U.S.A.**

01/2017 – 05/2020 • Ph.D. in Industrial Engineering
Dissertation: *Economic Feasibility of Adopting Additive Manufacturing for Low-volume Demand*
Committee: *Mingzhou Jin (advisor), Sean Willems, Xueping Li, James Ostrowski*

08/2015 – 12/2016 • M.S. in Industrial Engineering
Thesis: *A Mixed Integer Linear Programming Approach for Developing Salary Administration Systems*
Committee: *Alberto Garcia-Diaz (advisor), John Kobza, Rapinder Sawhney*

***The University of Georgia*, Athens, GA, U.S.A.**

08/2014 – 08/2015 • Non-degree Intensive English Program at the English Language Institute

***Nevsehir Haci Bektas Veli University*, Nevsehir, Türkiye**

09/2008 – 06/2012 • B.S. in Business Administration

Employment History

***Argonne National Laboratory*, Lemont, IL, U.S.A.**

07/2024 – ····· • Computational Transportation Engineer
05/2021 – 07/2024 • Research Consultant
05/2020 – 05/2021 • Postdoctoral Appointee Transportation System Modeler
01/2020 – 05/2020 • Visiting Student (Consultant)
05/2019 – 08/2019 • W. J. Cody Associate Intern

***Tarsus University*, Mersin, Türkiye**

11/2021 – 09/2022 • Assistant Professor
03/2021 – 11/2021 • Lecturer

***Toros University*, Mersin, Türkiye**

03/2021 – 11/2021 • Lecturer

***The University of Tennessee*, Knoxville, TN, U.S.A.**

03/2017 – 05/2020 • Graduate Research/Teaching Assistant
08/2015 – 03/2017 • Note Taker & Test Proctor at the Student Disability Services

***Republic of Türkiye Ministry of National Education*, Ankara, Türkiye**

12/2013 – 05/2020 • Government Sponsored Scholar (through YLSY program)

***Oak Ridge National Laboratory*, Oak Ridge, TN, U.S.A.**

05/2018 – 08/2018 • Advanced Short-Term Research Opportunity (ASTRO) Intern

***Kuwait-Turkish Participation Bank*, Adana, Türkiye**

08/2012 – 07/2014 • Retail and Small Business Sales Assistant

***Vakifbank*, Adana, Türkiye**

06/2011 – 08/2011 • Operations and Accounting Intern

***Six Flags Darien Lake*, Buffalo, NY, U.S.A. & *Great Adventure*, Jackson, NJ, U.S.A.**

06/2009 – 09/2009 • Sales Representative

Honors and Awards

Professional & Academic Honors

2024 • *Impact Argonne Award (Extraordinary Effort)* – Argonne National Laboratory

2018 • *Best Track Paper Award* – Logistics & Supply Chain Division of IISE for the conference paper: "An Agent-Based Simulation Model for Optimization of Spare Parts Sourcing with Additive Manufacturing" by Cokyasar T., Kose H., Dong W., Li X., Jin M., and Li W.

2018 • *2nd Best Track Paper Award* – Engineering Management Division of IISE for the conference paper: "Optimization of Maintenance and Infrastructure Projects Based on Risk Factors" by Hassler M., Cokyasar T., Jin M., and Tang R.

2018 • *Extraordinary Professional Promise Award* – Chancellor's Citation Awards, the University of Tennessee

2017 • *Service Recognition Award* – Student Disability Services, the University of Tennessee

Fellowships & Scholarships

2019 – 2020 • *E.J. Sierleja Memorial Fellowship* – IISE

2017 – 2019 • *Helen Jubin Fellowship Award* – College of Engineering, the University of Tennessee

2013 – 2020 • *Graduate Fellowship Award* – Republic of Türkiye Ministry of National Education

2008 – 2012 • *Undergraduate Fellowship Award* – Renaissance Education Foundation (Rönesans Eğitim Vakfı)

Projects

Argonne National Laboratory, Lemont, IL, U.S.A.

05/2021 – 05/2025 • *Principle Investigator* – Optimization for Transit, Freight, and Urban Mobility Problems, U.S. Department of Energy through Argonne National Laboratory ($450,000)

2020 – 2021 • *Postdoctoral Researcher* – Department of Energy (DOE) High Performance Computing for Mobility (HPC4Mobility): Chicago Transit Authority Transit Network Efficiency and the Changing Mobility Landscape ($25,000)

2020 – 2021 • *Postdoctoral Researcher* – Department of Energy (DOE) Systems & Modeling for Accelerated Research in Transportation (SMART) Mobility 2.0 ($50,000)

01/2020 – 05/2020 • *Graduate Researcher* – Enhancing Polaris Agent Based Framework ($60,000)

05/2019 – 08/2019 • *Graduate Researcher* – Optimal Assignment for the Single household Shared Autonomous Vehicle ($20,000)

Oak Ridge National Laboratory, Oak Ridge, TN, U.S.A.

05/2018 – 08/2018 • *Graduate Researcher* – Optimization for Smart Manufacturing Technology Adoption ($15,000)

Publications and Conferences

Journal Articles (*Corresponding Author)

1. Hosseini Dolatabadi, S. H., Bhuiyan, T. H., Kaleem, W., Subramanyam, A., & **Cokyasar**, **T.** (2026). A branch-and-cut algorithm for routing a heterogeneous drone-ground vehicle fleet to deliver time-sensitive products. *Transportation Research Part C: Emerging Technologies*, *183*. https://doi.org/10.1016/j.trc.2025.105454
2. Ismael, A., & **Cokyasar**, **T.** (2026). Modeling and calibration of the supplier selection problem in freight agent–based simulations. *Transportation Research Record*. https://doi.org/10.1177/03611981261455027
3. Kaleem, W., **Cokyasar***, **T.**, Larson, J., Verbas, O., Bhuiyan, T. H., & Subramanyam, A. (2026). Extreme-scale EV charging infrastructure planning for last-mile delivery using high-performance parallel computing [In Press]. *Transportation Research Part B: Methodological*, *205*. https://doi.org/10.1016/j.trb.2026.103403
4. Ng, T., Max, Mahmassani, H., Tong, D., Verbas, O., & **Cokyasar**, **T.** (2026). Joint optimization of multimodal transit frequency and shared autonomous vehicle fleet size with hybrid metaheuristic and nonlinear programming. *Transportation Research Part C: Emerging Technologies*, *185*, 105568. https://doi.org/10.1016/j.trc.2026.105568

5. Verbas, O., **Cokyasar**, **T.**, Joyce-Johnson, S., Wainwright, S., Coates, M., Rousseau, A., Aloisi, J., Stewart, A., & Auld, J. (2026). Impact of transit on mobility, equity, and economy in the chicago metropolitan region [Accepted]. *Public Transport*. https://arxiv.org/pdf/2409.04568
6. Bazarnovi, S., **Cokyasar**[*], **T.**, Verbas, O., & Mohammadian, A. K. (2025). Problem of locating and allocating charging equipment for battery electric buses under stochastic charging demand. *European Journal of Operational Research*. https://doi.org/10.1016/j.ejor.2025.07.064
7. Paul, J., Gurumurthy, K. M., **Cokyasar**, **T.**, Su, H., Khan, N., Auld, J., & Jia, Y. (2025). Proactive assignment strategy with human choice models for boosting pooled rideshare service. *IEEE Transactions on Intelligent Transportation Systems*, *26*, 2351–2365. https://doi.org/10.1109/TITS.2024.3515074
8. Davatgari, A., **Cokyasar**[*], **T.**, Subramanyam, A., Larson, J., & Mohammadian, A. K. (2024). Electric vehicle supply equipment location and capacity allocation problem. *European Journal of Operational Research*, *317*, 953–966. https://doi.org/10.1016/j.ejor.2024.04.022
9. Davatgari, A., **Cokyasar**, **T.**, Verbas, O., & Mohammadian, A. K. (2024). Heuristic solutions to the single depot electric vehicle scheduling problem with next day operability constraints. *Transportation Research Part C: Emerging Technologies*, *163*, 104656. https://doi.org/10.1016/j.trc.2024.104656
10. Dean, M. D., Gurumurthy, K. M., Zhou, Z., Verbas, O., **Cokyasar**, **T.**, & Kockelman, K. (2024). An integrated transportation-power system model for a decarbonizing world. *Transportation Planning and Technology*, *47*, 1277–1313. https://doi.org/10.1080/03081060.2024.2348721
11. Ng, M. T. M., Mahmassani, H. S., Verbas, O., **Cokyasar**, **T.**, & Engelhardt, R. (2024). Redesigning large-scale multimodal transit networks with shared autonomous mobility services. *Transportation Research Part C: Emerging Technologies*, *168*, 104575. https://doi.org/10.1016/j.trc.2024.104575
12. **Cokyasar**[*], **T.**, Stinson, M., Sahin, O., Prabhakar, N., & Karbowski, D. (2023). Comparing regional energy consumption for direct drone and truck deliveries. *Transportation Research Record*, *2677*(2), 310–327. https://doi.org/10.1177/03611981221145137
13. **Cokyasar**[*], **T.**, de Souza, F., Auld, J., & Verbas, O. (2022). Dynamic ride-matching for large-scale transportation systems. *Transportation Research Record*, *2676*(3), 172–182. https://doi.org/10.1177/03611981211049422
14. **Cokyasar**, **T.**, & Jin, M. (2022). Additive manufacturing capacity allocation problem over a network. *IISE Transactions*, *55*(8), 807–820. https://doi.org/10.1080/24725854.2022.2120222
15. **Cokyasar**[*], **T.**, Subramanyam, A., Larson, J., Stinson, M., & Sahin, O. (2022). Time-constrained capacitated vehicle routing problem in urban e-commerce delivery. *Transportation Research Record*, *2677*(2), 190–203. https://doi.org/10.1177/03611981221124592
16. Subramanyam, A., **Cokyasar**, **T.**, Larson, J., & Stinson, M. (2022). Joint routing of conventional and range-extended electric vehicles in a large metropolitan network. *Transportation Research Part C: Emerging Technologies*, *144*, 103830. https://doi.org/10.1016/j.trc.2022.103830
17. **Cokyasar**[*], **T.** (2021). Optimization of battery swapping infrastructure for e-commerce drone delivery. *Computer Communications*, *168*, 146–154. https://doi.org/10.1016/j.comcom.2020.12.015
18. **Cokyasar**[*], **T.**, Dong, W., Jin, M., & Verbas, İ. Ö. (2021). Designing a drone delivery network with automated battery swapping machines. *Computers & Operations Research*, *129*, 105177. https://doi.org/10.1016/j.cor.2020.105177
19. **Cokyasar**, **T.**, & Larson, J. (2020). Optimal assignment for the single-household shared autonomous vehicle problem. *Transportation Research Part B: Methodological*, *141*, 98–115. https://doi.org/10.1016/j.trb.2020.09.003
20. **Cokyasar**, **T.**, Garcia-Diaz, A., & Jin, M. (2019). Optimization of size and timing of base salary increases. *The Engineering Economist*, *64*(2), 97–115. https://doi.org/10.1080/0013791X.2018.1528408

In Progress and Submitted Manuscripts

1. Alam, M. R., Verbas, O., **Cokyasar**, **T.**, Bhidya, H., Altman, J., Beck, N., Auld, J., Veiga de Camargo, P., Cook, J., de Souza, F., Nair, G., Uhm, H., & Zill, J. (n.d.). Modeling of mobility and energy policies in an agent-based framework: Case studies for Chicago region in 2050. Submitted to the *106th Transportation Resarch Board Annual Meeting (Washington, DC, US, 01/10/2027 - 01/14/2027)*.

2. Bazarnovi, S., **Cokyasar**, **T.**, Verbas, O., & Mohammadian, A. (n.d.). Integrated optimization of scheduling and flexible charging in mixed electric-diesel urban transit bus systems. Submitted to the *Transportation Research Part E: Logistics and Transportation Review* [Under Review]. https://arxiv.org/pdf/2601.11751
3. Bhidya, H., **Cokyasar**, **T.**, & Verbas, O. (n.d.). Bus fleet electrification under capital cost and scheduling constraints: A five-agency case study. Submitted to the *106th Transportation Resarch Board Annual Meeting (Washington, DC, US, 01/10/2027 - 01/14/2027)*.
4. **Cokyasar**, **T.** (n.d.). Intercity EV charging readiness in Türkiye: A countrywide scheduling and economic assessment. Submitted to the *Journal of Turkish Operations Management*. https://arxiv.org/pdf/2607.25143
5. **Cokyasar**, **T.** (n.d.). A national-scale ev charging scheduling framework: Optimal detour routing under infrastructure capacity constraints. Submitted to the *106th Transportation Resarch Board Annual Meeting (Washington, DC, US, 01/10/2027 - 01/14/2027)*. https://arxiv.org/pdf/2607.18453
6. Hajibagher Najjar, Z., **Cokyasar**, **T.**, Alam, M., & Mohammadian, A. K. (n.d.). Joint fleet sizing, planning, and electrification for last-mile delivery: A two-stage continuous approximation and optimization approach. Submitted to the *106th Transportation Resarch Board Annual Meeting (Washington, DC, US, 01/10/2027 - 01/14/2027)*.
7. Kaleem, W., **Cokyasar**, **T.**, Ismael, A., Sahin, O., & Uhm, H. S. (n.d.). Approximate dynamic programming for matching, scheduling, and pricing in truckload freight markets. Submitted to the *106th Transportation Resarch Board Annual Meeting (Washington, DC, US, 01/10/2027 - 01/14/2027)*.
8. Ng, T., Max, Raymer, M., Mahmassani, H., Verbas, O., & **Cokyasar**, **T.** (n.d.). Equity impacts of public transit network redesign with shared autonomous mobility services. Submitted to the *Public Transport* - [Under Review]. https://arxiv.org/abs/2501.01615
9. Paul, J., Su, H., Gurumurthy, K. M., **Cokyasar**, **T.**, Auld, J., & Jia, Y. (n.d.). A study of financial impacts of pooled rideshare based on assignment strategies. Submitted to the *Transportation Research Part A: Policy and Practice* [Under Review]. https://papers.ssrn.com/sol3/papers.cfm?abstract_id=5101649
10. Paul, J., Vankadaru, V., Gurumurthy, K. M., **Cokyasar**, **T.**, Auld, J., & Jia, Y. (n.d.). Strategic deployment of tncs for sustainable urban transportation: Insights from Austin. Submitted to the *IEEE Transactions on Intelligent Transportation Systems* - [Under Review].
11. Vakayil, A., de Souza, F., **Cokyasar**, **T.**, Gurumurthy, K. M., & Larson, J. (n.d.). Large-scale dynamic ridesharing with iterative assignment. https://arxiv.org/abs/2311.06160
12. Verbas, O., **Cokyasar**, **T.**, de Camargo, P. V., Gurumurthy, K. M., Zuniga-Garcia, N., & Auld, J. (n.d.). Modeling transit in a fully integrated agent-based framework: Methodology and large-scale application. Submitted to the *Public Transport* - [Under Review]. https://arxiv.org/pdf/2408.05176
13. Wu, Y., **Cokyasar**, **T.**, Jayashankar, N., Gurumurthy, K. M., & Larson, J. (n.d.). Multi-household shared autonomous vehicle problem with subscription tiers. Submitted to the *106th Transportation Resarch Board Annual Meeting (Washington, DC, US, 01/10/2027 - 01/14/2027)*.

Conference Papers

1. Acosta, J. G., Verbas, O., **Cokyasar**, **T.**, Auld, J., Altman, J., Beck, N., Alhajjar, M., de Camargo, P. V., Cook, J., & Derrible, S. (2026). Agent-based modeling of transportation energy shifts and trends: A case study of the Chicago region. 605.
2. Bazarnovi, S., **Cokyasar**, **T.**, Verbas, O., & Mohammadian, A. K. (2025). Optimizing charging infrastructure for battery electric buses: A stochastic location model with budget constraints. In: *The 16th International Conference on Ambient Systems, Networks and Technologies (ANT), Procedia Computer Science, 257*, 583–590.
3. **Cokyasar**, **T.**, Bazarnovi, S., Verbas, O., & Mohammadian, A. K. (2025). Assessing the impact of route interlining via bus schedule optimization. In: *The 16th International Conference on Emerging Ubiquitous Systems and Pervasive Networks (EUSPN), Procedia Computer Science* [Accepted].

4. Davatgari, A., Mohammadi, M. Y., **Cokyasar**, **T.**, Mohammadian, A. K., & Auld, J. (2024). A label-correcting algorithm for constrained one-to-many k-shortest path problem with replenishment. In: *The 15th International Conference on Ambient Systems, Networks and Technologies (ANT), Procedia Computer Science, 238*, 369–376. https://doi.org/10.1016/j.procs.2024.06.037
5. Davatgari, A., Verbas, O., **Cokyasar**, **T.**, & Mohammadian, A. K. (2024). Optimization of electric bus scheduling and charger location. In: *The 15th International Conference on Ambient Systems, Networks and Technologies (ANT), Procedia Computer Science, 238*, 361–368. https://doi.org/10.1016/j.procs.2024.06.036
6. **Cokyasar**, **T.**, Davatgari, A., & Mohammadian, A. K. (2023). An optimization model for solving the route clustering problem. In: *The 14th International Conference on Ambient Systems, Networks and Technologies (ANT), Procedia Computer Science, 220*, 180–186. https://doi.org/10.1016/j.procs.2023.03.025
7. **Cokyasar**, **T.**, Verbas, O., & Auld, J. (2023). Electric vehicle scheduling problem with tour combinations. In: *The 14th International Conference on Ambient Systems, Networks and Technologies (ANT), Procedia Computer Science, 220*, 413–420. https://doi.org/10.1016/j.procs.2023.03.053
8. **Cokyasar**, **T.**, Verbas, O., Davatgari, A., & Mohammadian, A. K. (2023). Solving the electric vehicle scheduling problem at large-scale. In: *IEEE 26th International Conference on Intelligent Transportation Systems (ITSC)*, 1134–1139. https://doi.org/10.1109/ITSC57777.2023.10422441
9. Paul, J., Gurumurthy, K. M., **Cokyasar**, **T.**, Su, H., Auld, J., & Jia, Y. (2023). Optimization of dynamic ride-sharing by considering user preference through discount and delay tolerance. In: *IEEE 26th International Conference on Intelligent Transportation Systems (ITSC)*, 2770–2775. https://doi.org/10.1109/ITSC57777.2023.10422612
10. **Cokyasar**, **T.** (2021). Delivery drone route planning over a battery swapping network. In: *The 12th International Conference on Ambient Systems, Networks and Technologies (ANT), Procedia Computer Science, 184*, 10–16. https://doi.org/10.1016/j.procs.2021.03.013
11. **Cokyasar**, **T.**, Auld, J., Javanmardi, M., Verbas, O., & de Souza, F. (2020). Analyzing energy and mobility impacts of privately-owned autonomous vehicles. In: *IEEE 23rd International Conference on Intelligent Transportation Systems (ITSC)*, 1–6. https://doi.org/10.1109/ITSC45102.2020.9294218
12. **Cokyasar**, **T.**, Dong, W., & Jin, M. (2019). Network optimization for hybrid last-mile delivery with trucks and drones. In: *IISE Annual Conference Proceedings*, 1250–1255, ISBN: 978-0-9837624-8-5.
13. **Cokyasar**, **T.**, Kose, H., Dong, W., Li, X., Jin, M., & Li, W. (2018). An agent-based simulation model for optimization of spare parts sourcing with additive manufacturing. In: *IISE Annual Conference Proceedings*, 982–987, ISBN: 978-1- 5108-6935-6.
14. Hassler, M., **Cokyasar**, **T.**, Jin, M., & Tang, R. (2018). Optimization of maintenance and infrastructure projects based on risk factors. In: *IISE Annual Conference Proceedings*, 624–629, ISBN: 978-1-5108-6935-6.
15. **Cokyasar**, **T.**, Garcia-Arcaya, C., & Garcia-Diaz, A. (2017). A mixed integer linear programming MATLAB-based methodology for the development of a salary administration guide. In: *IISE Annual Conference Proceedings*, 506–511, ISBN: 978-0-9837624-6-1.

Conference Presentations

1. Acosta, J., Verbas, O., **Cokyasar**, **T.**, Auld, J., Altman, J., Beck, N., Alhajjar, M., Camargo, P., Cook, J., & Derrible, S. (2026). Agent-based modeling of transportation energy shifts and trends: A case study of the chicago region. *105th Transportation Research Board Annual Meeting*. Poster was jointly presented by Verbas, O. and Acosta, J. in Washington, DC, January 10-15.
2. Bassaleh, A., **Cokyasar**, **T.**, & Yildirim, M. B. (2026). Trajectory-based electric vehicle charging infrastructure planning *INFORMS Annual Meeting*. Abstract will be presented by Bassaleh, A. in San Francisco, CA, November 1-4.
3. Bazarnovi, S., **Cokyasar**, **T.**, Verbas, O., & Mohammadian, A. K. (2026). Integrated optimization of scheduling and flexible charging in mixed electric-diesel urban transit bus systems. *INFORMS Annual Meeting*. Abstract will be presented by Cokyasar, T. in San Francisco, CA, November 1-4.

4. Bazarnovi, S., **Cokyasar**, **T.**, Verbas, O., & Mohammadian, A. K. (2026). A unified framework for fleet composition, vehicle scheduling, and charging in electric diesel transit systems. *105th Transportation Research Board Annual Meeting*. Poster was presented in Washington, DC, January 10-15.
5. **Cokyasar**, **T.**, Bazarnovi, S., Verbas, O., Ismael, A., & Mohammadian, A. K. (2026). Optimizing bus fleet utilization through router interlining: A multi-agency case study. *105th Transportation Research Board Annual Meeting*. Poster was jointly presented by Verbas, O. and Cokyasar, T. in Washington, DC, January 10-15.
6. Hajibagher Najjar, Z., **Cokyasar**, **T.**, & Mohammadian, A. K. (2026). Multi-year fleet replacement and electrification planning for last-mile delivery with routing-based cost and mileage approximations. *INFORMS Annual Meeting*. Abstract will be presented by Najjar, Z. in San Francisco, CA, November 1-4.
7. Hosseini Dolatabadi, S. H., Bhuiyan, T. H., Kaleem, W., Subramanyam, A., & **Cokyasar**, **T.** (2026). A branch-and-cut algorithm for routing a heterogeneous drone-ground vehicle fleet to deliver time-sensitive products *INFORMS Annual Meeting*. Abstract will be presented by Hosseini Dolatabadi, H. in San Francisco, CA, November 1-4.
8. Ismael, A., & **Cokyasar**, **T.** (2026). From optimization to national scale: A two-stage framework for calibrated supplier selection in freight agent-based models. *INFORMS Annual Meeting*. Abstract will be presented by Ismael, A. in San Francisco, CA, November 1-4.
9. Ismael, A., & **Cokyasar**, **T.** (2026). Modeling and calibration of supplier selection problem in freight agent-based simulations. *105th Transportation Research Board Annual Meeting*. Study was presented in Washington, DC, January 10-15.
10. Paul, J., **Cokyasar**, **T.**, Gurumurthy, K. M., Auld, J., & Jia, Y. (2026). Improving experience beyond the ride: Joint assignment and repositioning for rideshare. *105th Transportation Research Board Annual Meeting*. Poster was presented in Washington, DC, January 10-15.
11. Bazarnovi, S., **Cokyasar**, **T.**, Verbas, O., & Mohammadian, A. K. (2025). Strategic allocation of battery electric bus chargers under stochastic demand: A chicago case study using metaheuristic approaches. *104th Transportation Research Board Annual Meeting*. Poster was presented by Bazarnovi, S. in Washington, DC, January 5-9.
12. Borlaug, B., Sahin, O., Yip, A., Sun, J., **Cokyasar**, **T.**, Mansour, C., Jadun, P., & Muratori, M. (2025). Highly resolved transportation electricity load projections for the united states. *104th Transportation Research Board Annual Meeting*. Poster was presented by Borlaug, B. in Washington, DC, January 5-9.
13. Ismael, A., Sahin, O., Uhm, H.-s., Shen, H., Zuniga-Garcia, N., Huang, Y., **Cokyasar**, **T.**, Gurumurthy, K. M., Auld, J., & Stinson, M. (2025). Integration and application of a freight agent-based simulation framework. *104th Transportation Research Board Annual Meeting*. Poster was presented by Ismael, A. in Washington, DC, January 5-9.
14. Kaleem, W., **Cokyasar**, **T.**, Larson, J., Subramanyam, A., Verbas, O., & Bhuiyan, T. H. (2025). Large scale stochastic charger location and allocation problem in last-mile delivery . *104th Transportation Research Board Annual Meeting*. Poster was presented by Kaleem, W. in Washington, DC, January 5-9.
15. Ng, M., Tong, D., Mahmassani, H., Verbas, O., & **Cokyasar**, **T.** (2025). Joint optimization of pattern, headway, and fleet size of multiple urban transit lines with perceived headway consideration and passenger flow allocation. *104th Transportation Research Board Annual Meeting*. Poster was presented by Ng, M. in Washington, DC, January 5-9.
16. Paul, J., Su, H., Gurumurthy, K. M., **Cokyasar**, **T.**, Auld, J., & Jia, Y. (2025). A study of financial impacts of pooled rideshare based on assignment strategies. *104th Transportation Research Board Annual Meeting*. Poster was presented by Paul, J. in Washington, DC, January 5-9.
17. Sahin, O., Borlaug, B., **Cokyasar**, **T.**, Zuniga-Garcia, N., & Mansour, C. (2025). Analysis of load profiles for medium and heavy-duty electric vehicles: Implications for grid integration. *104th Transportation Research Board Annual Meeting*. Poster was presented by Sahin, O. in Washington, DC, January 5-9.
18. Verbas, O., **Cokyasar**, **T.**, Camargo, P. V. d., Gurumurthy, K. M., Zuniga-Garcia, N., & Auld, J. (2025). Modeling transit in a fully integrated agent-based framework: Methodology and large-scale application. *104th Transportation Research Board Annual Meeting*. Poster was presented by Verbas, O. in Washington, DC, January 5-9.

19. Verbas, O., **Cokyasar**, **T.**, Joyce-Johnson, S., Wainwright, S., Coates, M., Rousseau, A., Aloisi, J., Stewart, A., & Auld, J. (2025). Impact of transit on mobility, equity, and economy in the chicago metropolitan region. *104th Transportation Research Board Annual Meeting*. Poster was presented by Verbas, O. in Washington, DC, January 5-9.
20. Bazarnovi, S., **Cokyasar**, **T.**, Verbas, O., & Mohammadian, A. K. (2024). Locating and allocating battery electric bus chargers with stochastic charging demand. *INFORMS Annual Meeting*. Abstract was presented by Bazarnovi, S. in Seattle, WA, October 20-23.
21. **Cokyasar**, **T.**, Verbas, O., & Bhuyain, T. (2024). Optimizing e-commerce electric delivery vehicle charger locations under stochastic demand. *INFORMS Annual Meeting*. Abstract will be presented in Seattle, WA, October 20-23.
22. Davatgari, A., **Cokyasar**, **T.**, Verbas, O., & Mohammadian, A. K. (2024). Heuristic approaches for the electric vehicle scheduling problem: Large-scale application with next day operability constraints. *103rd Transportation Research Board Annual Meeting*. Abstract was presented, Jan. 7-11.
23. Davatgari, A., Mohammadi, M. Y., **Cokyasar**, **T.**, & Mohammadian, A. K. (2024). A label-correcting algorithm for constrained one-to-many k-shortest path problem with replenishment. *103rd Transportation Research Board Annual Meeting*. Abstract was presented by Davatgari, A., Jan. 7-11.
24. Dean, M. D., Gurumurthy, K. M., Zhou, Z., Verbas, O., **Cokyasar**, **T.**, & Kockelman, K. (2024). An integrated transportation-power system model for a decarbonizing world. *103rd Transportation Research Board Annual Meeting*. Abstract was presented by Dean, M., Jan. 7-11.
25. Huang, Y., Verbas, O., Gurumurthy, K. M., Sahin, O., **Cokyasar**, **T.**, Zuniga, N., & Auld, J. (2024). Future transportation electrification: Operations and impacts . *Intelligent Transportation Systems America Conference and Expo*. Abstract was presented by Huang, Y. in Phoenix, AZ, May 25.
26. Ng, M., Mahmassani, H., Verbas, O., **Cokyasar**, **T.**, & Tong, D. (2024). Train service design problem with perceived headwayconsideration and passenger flow allocation for urbanrail transit lines. *INFORMS Annual Meeting*. Abstract was presented in Seattle, WA, October 20-23.
27. Ng, M. T. M., Raymer, M., Mahmassani, H. S., Verbas, O., & **Cokyasar**, **T.** (2024). Can redesigning public transit network with shared autonomous mobility services improve equity? *103rd Transportation Research Board Annual Meeting*. Abstract was presented by Ng, M., Jan. 7-11.
28. Ng, M. T. M., Raymer, M., Mahmassani, H. S., Verbas, O., & **Cokyasar**, **T.** (2024). Spatial and vertical equity impacts of public transit network redesign with shared autonomous mobility services. *Transit Data 2024: The 9th International Workshop and Symposium on Research and Applications on the Use of Passive Data from Public Transport*. Abstract was presented by Ng, M. in London, UK, July 01-04.
29. Paul, J., Gurumurthy, K. M., **Cokyasar**, **T.**, Su, H., Jia, Y., & Auld, J. (2024). Integrating human factors in dynamic rideshare assignment: Willingness-to-pay for delay. *103rd Transportation Research Board Annual Meeting*. Abstract was presented by Paul, J., Jan. 7-11.
30. Verbas, O., & **Cokyasar**, **T.** (2024). Charger location and scheduling optimization for battery electric buses. *INFORMS International Meeting*. Abstract was presented by Verbas, O. in Medellin, Colombia, June 16-19.
31. Verbas, O., **Cokyasar**, **T.**, Rousseau, A., Auld, J., Joyce-Johnson, S., Aloisi, J., Stewart, A., & Barrett, G. (2024). Mobility, equity, and economic impact of transit in the Chicago metropolitan region. *INFORMS Annual Meeting*. Abstract will be presented in Seattle, WA, October 20-23.
32. **Cokyasar**, **T.**, Verbas, O., Davatgari, A., & Mohammadian, A. K. (2023). A heuristic solution to the single depot electric vehicle scheduling problem. *INFORMS Annual Meeting*. Abstract was presented by Verbas, O. in Phoenix, AZ, Oct. 15-18.
33. Davatgari, A., **Cokyasar**, **T.**, Subramanyam, A., Larson, J., & Mohammadian, A. K. (2023). Electric vehicle supply equipment allocation for fixed route vehicles. *IISE Annual Meeting & Expo*. Abstract was presented in New Orleans, LA, May 19-23.
34. Davatgari, A., **Cokyasar**, **T.**, Subramanyam, A., Larson, J., & Mohammadian, A. K. (2023). Locating and allocating chargers for fixed-route freight delivery. *INFORMS Annual Meeting*. Abstract was presented in Phoenix, AZ, Oct. 15-18.

35. Davatgari, A., **Cokyasar**, **T.**, Subramanyam, A., & Stinson, M. (2023). Electric vehicle supply equipment location and allocation for freight electrification. *Transportation Research Board Annual Meeting*. Poster presented by Davatgari, A. in Washington, DC, January 8-12.
36. Dean, M. D., Gurumurthy, K. M., Zhou, Z., Verbas, O., **Cokyasar**, **T.**, & Kockelman, K. (2023). An integrated transportation-power system model for a decarbonizing world. *Bridging Transportation Researchers (BTR) Online Conference*. Abstract was presented online by Dean, M., Aug. 9-10.
37. **Cokyasar**, **T.** (2022). Last-mile drone and truck delivery energy comparisons. *INFORMS Annual Meeting*. Abstract was presented in Indianapolis, IN, October 16-19.
38. **Cokyasar**, **T.**, Larson, J., Stinson, M., Sahin, O., Verbas, O., Freyermuth, V., & Uhm, H. S. (2022). Truck routing optimization for large-scale transportation networks. *International Urban Freight Conference*. Abstract was presented by Uhm, H. S. in Long Beach, CA, May 25-27.
39. Subramanyam, A., **Cokyasar**, **T.**, Larson, J., & Stinson, M. (2022). Joint routing of conventional and range-extended electric vehicles in a large metropolitan network. *INFORMS Optimization Society Conference*. Abstract was presented by Subramanyam, A. in Greenville, SC, March 13-15.
40. Vakayil, A., Gurumurthy, K. M., de Souza, F., **Cokyasar**, **T.**, & Larson, J. (2022). Fast and scalable shared fleet algorithms for large-scale application. *INFORMS Annual Meeting*. Abstract was presented by Gurumurthy, K. M. in Indianapolis, IN, October 16-19.
41. Verbas, O., **Cokyasar**, **T.**, & Auld, J. (2022). Solving the electric vehicle scheduling problem at scale. *INFORMS Annual Meeting*. Abstract was presented by Verbas, O. in Indianapolis, IN, October 16-19.
42. **Cokyasar**, **T.** (2021). Planning routes for e-commerce delivery drones. *IISE Annual Meeting & Expo*. Abstract virtually presented.
43. Stinson, M., **Cokyasar**, **T.**, & Sahin, O. (2021). Mobility and sustainability impacts of advanced last-mile delivery technologies. *24th IEEE International Conference on Intelligent Transportation (ITSC2021) Last Mile Delivery and 50 Feet Logistics Challenges in the "New Normal" Workshop* Abstract was presented by Stinson, M. in Indianapolis, IN, September 19-22.
44. **Cokyasar**, **T.**, Dong, W., Jin, M., & Verbas, O. (2020). Designing a drone delivery network with automated battery swapping machines. *INFORMS Annual Meeting*. Abstract virtually presented as an invited speaker.
45. **Cokyasar**, **T.**, & Larson, J. (2020). Optimal assignment for the single-household shared autonomous vehicle problem. *IISE Annual Meeting & Expo*. Abstract virtually presented.
46. Verbas, O., **Cokyasar**, **T.**, & Auld, J. (2020). Chicago transit authority transit network efficiency and the changing mobility landscape. *U.S. Department of Energy's (DOE's) 2020 Annual Merit Review and Peer Evaluation Meeting (AMR) for the Hydrogen and Fuel Cells Program and the Vehicle Technologies Office, Poster Session EEMS088*. Poster virtually presented by Verbas, O.
47. **Cokyasar**, **T.**, Dong, W., & Jin, M. (2019). Network optimization for hybrid last mile delivery with trucks and drones. *INFORMS Computing Society*. Abstract was presented in Knoxville, TN.
48. **Cokyasar**, **T.**, Jin, M., & Willems, S. (2019). Spare parts supply chain optimization with additive manufacturing. *INFORMS Computing Society*. Abstract was presented in Knoxville, TN.
49. **Cokyasar**, **T.**, Jin, M., & Willems, S. (2018). Spare parts supply chain optimization with additive manufacturing. *INFORMS Annual Meeting*. Abstract was presented in Seattle, WA.

Professional Service

Editorial Service

2025 – · · · · • *Area Editor* – The Engineering Economist
2021 – 2022 • *Editorial Board Member* – International Journal of Applied Management Science (IJAMS)

Society Leadership

2025 – 2027 • *President* – IISE Engineering Economy Division
2023 – 2025 • *Director* – IISE Logistics and Supply Chain Division
2022 – 2024 • *Director* – IISE Engineering Economy Division
2019 – 2021 • *President* – IISE Logistics and Supply Chain Division Student Leadership Board
2018 – 2019 • *Vice President* – IISE Logistics and Supply Chain Division Student Leadership Board

Professional Service (continued)

Conference Organization

2026 • *Track Chair* – 2026 IISE Annual Meeting & Expo (Engineering Economy); the 17th International Conference on Emerging Ubiquitous Systems and Pervasive Networks (EUSPN – Adaptive Systems, Intelligent Computing and Applications)

2025 • *Track Chair* – 2025 IISE Annual Meeting & Expo (Engineering Economy & Facilities Design and Planning); the 16th International Conference on Emerging Ubiquitous Systems and Pervasive Networks (EUSPN – Adaptive Systems, Intelligent Computing and Applications)

2024 • *Track Chair* – 2024 IISE Annual Meeting & Expo (Engineering Economy); the 21st International Conference on Mobile Systems and Pervasive Computing (Mobi-SPC – Smart Cities and Ubiquitous Climate Change Management)

2023 • *Track Chair* – 2023 IISE Annual Meeting & Expo (Engineering Economy); the 14th International Conference on Emerging Ubiquitous Systems and Pervasive Networks (EUSPN – Adaptive Systems, Intelligent Computing and Applications)

Peer Review Activities

2016 – · · · · • *Ad-hoc Reviewer* – Transportation Research Part A: Policy and Practice (TRA), Transportation Research Part B: Methodological, Transportation Research Part C: Emerging Technologies (TRC), IISE Annual Meeting & Expo, IEEE Transactions on Engineering Management (TEM), IEEE Intelligent Transportation Systems Conference (ITSC), Computers & Operations Research (COR), Journal of Cleaner Production (JCLP), Transportation Research Board (TRB), Transportation Research Record (TRR), International Journal of Simulation and Process Modelling (IJSPM), Transportation Letters

Professional Memberships

2018 – · · · · • *Member* – INFORMS

2016 – · · · · • *Member* – IISE

Teaching Experience

Tarsus University, Mersin, Turkey

Spring 2022 • IE 104 Computer Programming I – Department of Industrial Engineering

Spring 2022 • IE 532 Advanced Computer Programming – Department of Industrial Engineering

Fall 2021 • IE 530 Linear Programming and Network Optimization – Department of Industrial Engineering

Toros University, Mersin, Turkey

Fall 2021 • INE 321 Production Planning – Department of Industrial Engineering

The University of Tennessee, Knoxville, TN, U.S.A.

Fall 2019 • IE 430/530 Advanced/Supply Chain Engineering – Department of Industrial & Systems Engineering, *Teaching Score*: 3.9/5.0

Fall 2018 • IE 430/530 Advanced/Supply Chain Engineering – Department of Industrial & Systems Engineering, *Teaching Score*: 4.2/5.0

Advising and Mentorship

2026 – Present • **Ahmad Bassaleh**, Research Collaborator (My role: Co-Mentor).
Wichita State University
Currently: Ph.D. Student.

2026 • **Yifan Wu**, Research Aide (My role: Primary Mentor).
Texas A&M University
Currently: Ph.D. Student.

2026 • **Annabelle Jung**, SULI Summer Intern (My role: Co-Mentor).
University of Illinois Chicago
Currently: B.S. Student.

Advising and Mentorship (continued)

2026 • **Waquar Kaleem**, Research Aide (My role: Primary Mentor).
Pennsylvania State University
Currently: Ph.D. Student.

2026 • **Hadi Bhidya**, SULI Spring Intern (My role: Primary Mentor).
University of Tennessee
Currently: B.S. Student.

2025 – Present • **Zahra Hajibagher Najjar**, Research Collaborator (My role: Primary Mentor).
University of Illinois Chicago
Currently: Ph.D. Student.

2024 – 2025 • **Draco Tong**, Research Collaborator (My role: Co-Mentor).
Northwestern University
Currently: Ph.D. Student.

2024 – 2025 • **Sadjad Bazarnovi, Ph.D.**, Research Collaborator (My role: Dissertation Co-Advisor).
University of Illinois Chicago.
Dissertation: *"Planning and Optimization of Charging Strategies for Electric Vehicles: From Infrastructure to Behavioral Insights."*
Currently: Operations Research Scientist at NetJets.

2024 • **Waquar Kaleem**, Givens Associate (My role: Co-Mentor).
Pennsylvania State University
Currently: Ph.D. Student.

2024 • **Selin Bayramoglu**, Research Aide (My role: Primary Mentor).
Georgia Institute of Technology.
Currently: Ph.D. Student.

2024 • **Joseph Paul, Ph.D.**, Research Aide (My role: Co-Mentor).
Clemson University.
Currently: Consultant at Plasman.

2023 – 2025 • **Max Ng, Ph.D.**, Research Collaborator (My role: Co-Mentor).
Northwestern University.
Currently: Machine Learning Engineer at Uber.

2023 • **Akhil Vakayil, Ph.D.**, Givens Associate (My role: Co-Mentor).
Georgia Institute of Technology.
Currently: Quantitative Analytics Specialist at Wells Fargo.

2022 – 2024 • **Amir Davatgari, Ph.D.**, Research Collaborator (My role: Dissertation Co-Advisor).
University of Illinois Chicago.
Dissertation: *"Medium-duty Electric Vehicle Infrastructure Planning and Operations in Urban Transportation Networks."*
Currently: Data Scientist at Starbucks.

Technical Skills

Programming • Python (pandas, NumPy, GeoPandas), R, Java, MATLAB, SQL
Optimization • Gurobi, CPLEX, PYOMO, AMPL, KNITRO, GLPK, IPOPT
Software & Tools • QGIS, Anylogic, SPSS, Git/GitHub, LaTeX
Databases • HDF5, SQLite
Languages • Turkish (Native), English (Fluent)